

IBRAHIM BOUBAKRI

Full Stack AI Developer | Backend & MLOps | Agentic Systems | DevOps

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Profile

Full stack AI developer and backend engineer experienced in modular architectures, Python/FastAPI services, data and execution pipelines, containerized deployment, CI/CD and observability. Builds reusable platform capabilities for AI and LLM applications, including agentic workflows, MCP-compatible tool integration, RAG-style retrieval, model abstraction, execution tracking and validation. Strong software-engineering foundation with a focus on reliability, automation, scalability and operational readiness.

Professional Experience

Full Stack AI Developer, Architect and DevOps

Modular AI Platform

Sep. 2024 – Present

Quebec City, QC, Canada / Remote

- Designed and developed a modular AI platform combining backend services, workflow orchestration, data management, observability, and AI/LLM integration.
- Built Python/FastAPI services and REST APIs for AI pipelines, structured data, metadata, execution logs, validation, and asynchronous processing.
- Designed an agentic execution layer that decomposes objectives into multi-step tasks, selects registered tools, invokes internal services through MCP-compatible interfaces / tool calling, preserves execution state, and validates outputs before continuing.
- Implemented autonomous workflows with structured outputs, retries/fallbacks, human-in-the-loop approval points, audit logs, and guardrail-style validation for safer execution.
- Added retrieval and grounding capabilities over platform artifacts and metadata using RAG, embeddings/vector search concepts, enabling agents and LLM workflows to use project context instead of relying only on prompts.
- Integrated local/open-source and external LLM services behind modular interfaces to support model substitution, routing, reproducible runs, and provider-independent application logic.
- Implemented Docker-based environments, automated tests, GitLab CI/CD pipelines, monitoring/logging, and deployment workflows; improved reliability through traceable execution and failure diagnostics.
- Used Claude Code, OpenAI Codex, GitHub Copilot and Cursor as supervised agentic engineering assistants for code analysis, refactoring, testing, documentation and incident resolution, with human review and CI validation.
- *Core technologies:* Python, FastAPI, REST APIs, MCP/tool integration, LLM workflows, RAG, SQL, Docker, GitLab CI/CD, PyTorch, OpenCV, Linux

Software Engineer, DevOps and Automation

PRIMATEC Engineering

Nov. 2023 – Aug. 2024

Sfax, Tunisia

- Developed Python tools and automation scripts that reduced manual work across software validation, testing, and quality-control processes.
- Integrated automated tests into CI/CD pipelines and maintained containerized continuous-integration environments using Docker, Jenkins and GitLab.
- Designed modular components with structured error handling, logging, traceability, and clear separation of responsibilities; produced installation and maintenance documentation.
- *Technologies:* Python, Docker, Jenkins, GitLab CI/CD, Jira, PowerShell, Batch, automated testing, Linux

Software Engineering Intern, Web Applications and Data

Hydrogen Research Institute

Mar. 2023 – Aug. 2023

Trois-Rivières, QC, Canada

- Designed and deployed a microservices-based web platform for real-time technical data collection, processing and visualization.
- Built frontend and backend services and automated ETL pipelines using MQTT; containerized services and contributed to CI/CD and Kubernetes-based deployment practices.
- *Technologies:* React, Node.js, MongoDB, Python, MQTT, Docker, Kubernetes, Jenkins

Researcher, Computer Vision and Autonomous Robotics

UQTR / Industry Collaboration with Noovel

2024 – 2026

Trois-Rivières, QC, Canada

- Designed an industrial computer-vision perception pipeline for pallet detection and geometric keypoint estimation from monocular RGB images.
- Built a Blender-based synthetic-data generation workflow with domain randomization and automatic annotations to support model training and transfer to real deployment environments.
- Developed an integrated AI and robotics platform connecting synthetic-data generation, model training, inference, PyBullet simulation, FastAPI services and a React interface.
- *Technologies:* Python, PyTorch, Ultralytics, CUDA, Blender, PyBullet, ROS 2, FastAPI, WebSocket, React, Jetson

Education

Université du Québec à Trois-Rivières, UQTR

Master's in Electrical Engineering, Computer Engineering Concentration

Sep. 2024 – Aug. 2026

Trois-Rivières, QC, Canada

National School of Computer Science, ENSI

Engineering Degree in Computer Science

Sep. 2020 – Aug. 2023

University of Manouba, Tunisia

Selected Projects

Modular Agentic AI Platform

Apr. 2026 – Present

- Designed a modular platform in which AI agents can plan multi-step work, discover and choose platform capabilities, call tools and APIs, and coordinate processing without hard-coding each end-to-end workflow.
- Defined an MCP-compatible tool/service layer for controlled access to platform functions such as data processing, file operations, validation, model execution and result generation.
- Implemented stateful agent workflows with structured inputs and outputs, task status, retries, fallbacks, auditability and human approval for sensitive or uncertain actions.
- Integrated RAG-style context retrieval, embeddings/vector-search concepts, model abstraction and response validation to improve grounding, reliability and portability across LLM providers.
- Instrumented runs with logs and execution metadata to support observability, debugging and evaluation of agent behaviour, latency and failures.
- *Technologies:* Python, FastAPI, Agentic AI, AI agents, MCP, tool/function calling, RAG, embeddings/vector search, REST APIs, SQL, Docker, CI/CD, LLM integration

HackZone Tunisia X | Infrastructure and Cybersecurity

Apr. 2022

- Helped organize a 24-hour capture-the-flag event for more than 100 teams and 300 participants; contributed to challenge infrastructure, service monitoring and incident resolution.
- Prepared technical challenges covering web vulnerabilities, Linux systems and reverse engineering.
- *Technologies:* Azure, Python, Kali Linux, Git, Docker, web cybersecurity

Technical Skills

AI and LLM Integration: reusable AI services, agent orchestration, MCP-compatible tool/service integration, model abstraction, RAG, LLM workflows, execution tracking, validation, observability

Backend Engineering: Python, FastAPI, Flask, Node.js, REST APIs, asynchronous processing, microservices, authentication, webhooks, error handling, logging

MLOps and DevOps: Docker, Kubernetes, GitLab CI/CD, Jenkins, Git, Linux, automated testing, deployment automation, monitoring, incident diagnostics

Data Platforms: SQL, MongoDB, MySQL, ETL, MQTT, metadata, structured files, data validation, pipelines

AI and ML: LLMs, Agentic AI, AI agents, tool calling, RAG, embeddings/vector-search concepts, PyTorch, OpenCV, response validation

Programming: Python, SQL, JavaScript, Java, PHP, C++, C#, PowerShell, Bash

Languages: French, B2 certified (TCF Canada, NCLC 7); English, professional working proficiency; Arabic, native

Certifications and Activities

- **DevOps Bootcamp** – TechWorld with Nana | **AWS Cloud Technical Essentials** – Coursera | **IBM DevOps and Software Engineering Professional Certificate** – in progress
- Organizing team member for ENSI Annual Forum, TuniHack, RoboCup and HackZone Tunisia.